Medical Imaging AI for Brain Tumor Diagnosis

GROUP NO: 9

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Introduction

Brain tumors are life-threatening conditions that require timely and accurate diagnosis for effective treatment.

Magnetic Resonance Imaging (MRI) is widely used for brain tumor detection, but manual analysis is time-consuming and prone to human error.

With increasing cases and limited clinical resources, there is a clear need for automated diagnostic support.

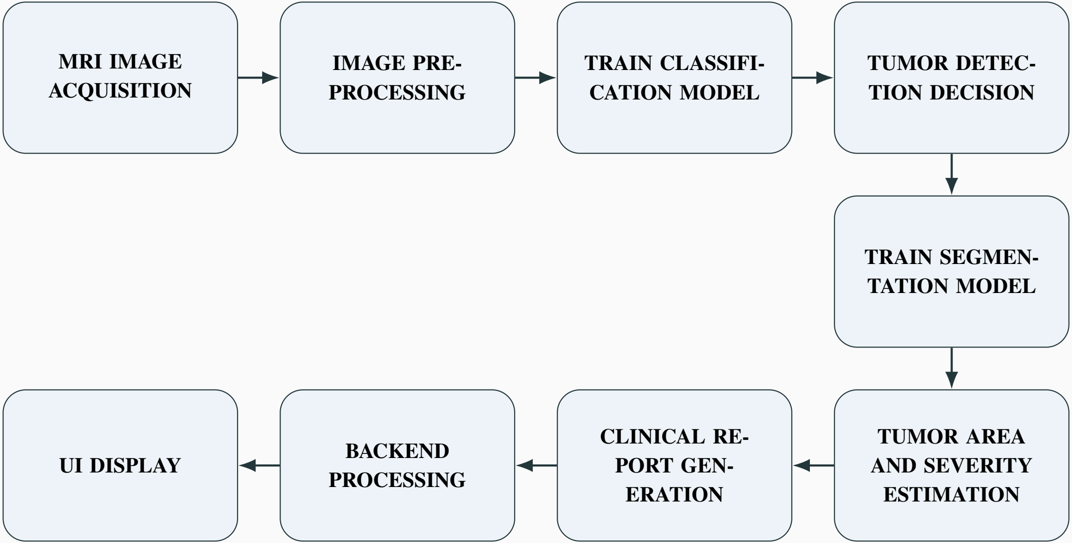
This project aims to develop an AI-based system using deep learning to classify brain MRI images as tumor or non-tumor.

In addition to classification, the system also generates a diagnostic report similar to that of a radiologist, providing comprehensive support for clinical decision-making.

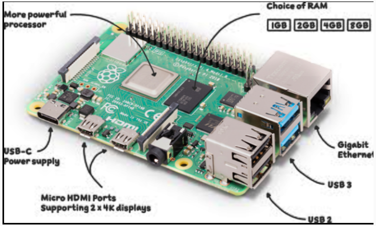
Objectives

- To develop a deep learning-based system for automated brain tumor diagnosis using MRI data.
- To implement transfer learning for achieving high accuracy and computational efficiency.
- To reduce diagnostic delays and assist radiologists in clinical decision-making.

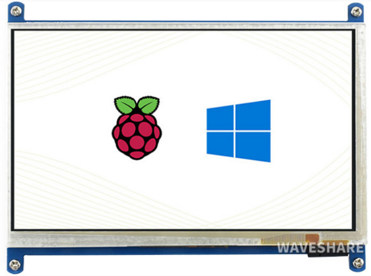
SYSTEM WORKFLOW



Hardware Components



Raspberry Pi 4 Model B



Touchscreen Display

Software Components

TensorFlow & TensorFlow Lite

PyTorch

ONNX Runtime

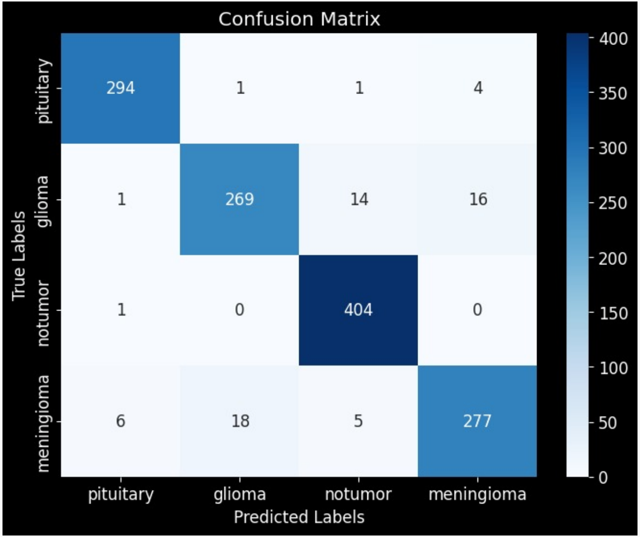
FastAPI

OpenCV

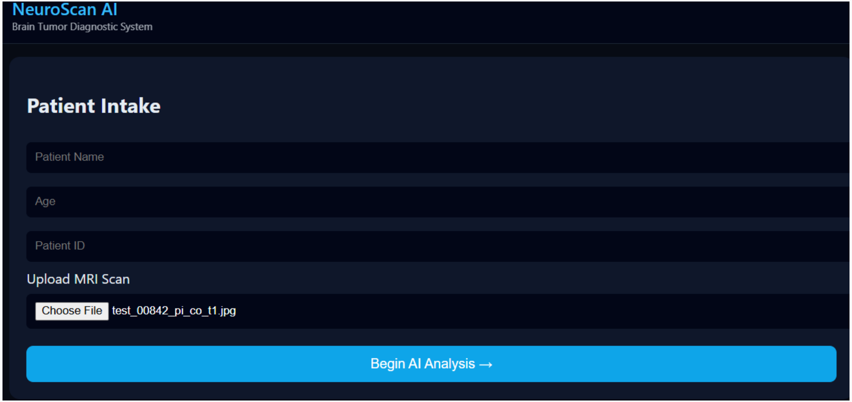
Work Done

- Developed a ResNet-50–based classification model for brain tumor detection and type identification
- Implemented a U-Net segmentation model for precise tumor localization and mask generation
- Integrated the classification and segmentation models into an end-to-end diagnostic pipeline with area and severity estimation
- Designed and deployed a web-based User Interface on Raspberry Pi using a FastAPI backend for report visualization

Confusion Matrix



Results

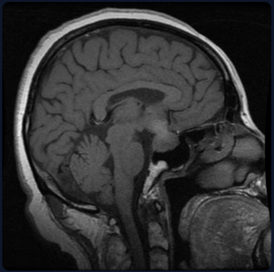

NeuroScan AI
Brain Tumor Diagnostic System
Patient Intake
Patient Name
Age
Patient ID
Upload MRI Scan
Choose File test_00842_pi_co_t1.jpg
Begin AI Analysis →

Results

NeuroScan AI

Brain Tumor Diagnostic System

Diagnosis Report



Tumor Type: pituitary

Confidence: 0.9996

Estimated Tumor Area: 0.81%

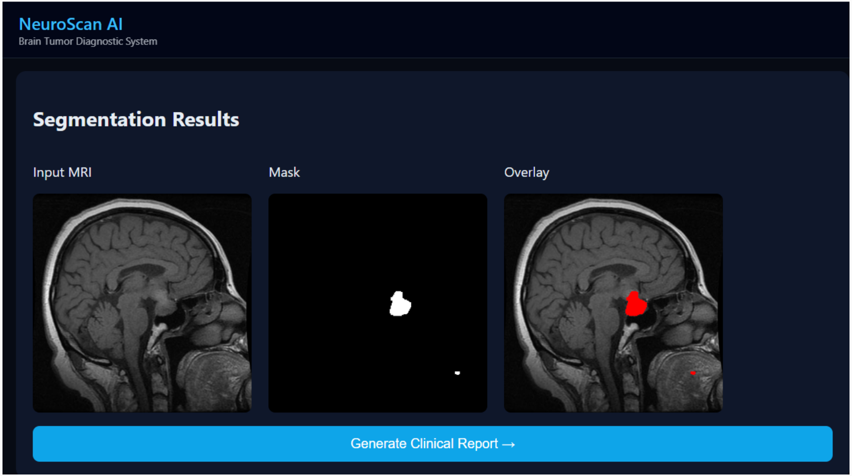
Severity: Mild

Interpretation

A small pituitary lesion is identified, likely representing a microadenoma.

View Segmentation →

Results


NeuroScan AI
Brain Tumor Diagnostic System
Segmentation Results
Input MRI
Mask
Overlay
Generate Clinical Report →

Results

NeuroScan AI

Brain Tumor Diagnostic System

Clinical Report

Name:

Age:

ID:

Findings

AI analysis indicates presence of a **pituitary** tumor with confidence **0.9996**. Estimated tumor area **0.81%**. Severity classified as **Mild**.

Interpretation

A small pituitary lesion is identified, likely representing a microadenoma.

Recommended Action

Further evaluation by a neurologist or neurosurgeon is advised. MRI with contrast and clinical correlation recommended.

AI HIGHLIGHTED TUMOR

Print Report

Segmentation Results

Conference Paper Reply

National Institute for Engineering & Research

ICROI-26

International Conference on Radiology and Oncologic Imaging (ICROI-26)

Date : 24-02-2026

Acceptance Letter

Dear Author(s)

We are pleased to inform you that your paper has been accepted by the review committee for Oral Presentation at the **International Conference on Radiology and Oncologic Imaging (ICROI-26)**

Paper ID: NIER_34982

Article Title: "Medical Imaging AI for Brain Tumor Diagnosis A Deep Learning-Based MRI Analysis System for Detection and Segmentation"

Author's Name: Adarsh A.S, Anjali Tomson Joseph, Anjana T, Anna Rose V Vinod

This conference will be held in **7th March 2026 on Delhi - India**

Current Limitations of the Project

- Limited dataset size may affect the model's ability to generalize to all real-world clinical cases.
- The system currently detects only selected tumor categories such as glioma, meningioma, pituitary tumor, and no tumor.
- The performance of the system depends on the quality and resolution of MRI images; noisy or low-quality scans may reduce accuracy.
- The deep learning models used in the system have limited explainability, making it difficult to clearly interpret how prediction results are generated.

Conclusion

- The proposed NeuroScan AI system provides an automated approach for brain tumor detection using MRI images.
- The system integrates ResNet50-based classification and deep learning based segmentation for accurate tumor analysis.
- The developed model is capable of detecting glioma, meningioma, pituitary tumor, and no tumor classes.
- Deployment on Raspberry Pi with a touchscreen interface enables a compact and portable embedded AI solution.
- The project demonstrates the potential of combining deep learning and embedded systems for intelligent medical diagnostic applications.

References (Part 1)

- B. Vinay Kumar *et al.*, “Building an Intelligent Brain Tumor System using Magnetic Resonance Imaging,” *Proc. 2023 Int. Conf. on Sustainable Computing and Data Communication Systems (ICSCDS)*, pp. 725–729, 2023. doi:10.1109/ICSCDS56580.2023.10105044
- K. Lata, P. Singh, S. Saini and L. R. Cenkeramaddi, ”Deep Learning-Based Brain Tumor Detection in Privacy-Preserving Smart Health Care Systems,” in IEEE Access, vol. 12, pp. 140722-140733, 2024, doi: 10.1109/ACCESS.2024.3456599.
- N. Noreen, S. Palaniappan, A. Qayyum, I. Ahmad, M. Imran and M. Shoaib, ”A Deep Learning Model Based on Concatenation Approach for the Diagnosis of Brain Tumor,” in IEEE Access, vol. 8, pp. 55135-55144, 2020, doi: 10.1109/ACCESS.2020.2978629.
- P. Khan et al., ”Machine Learning and Deep Learning Approaches for Brain Disease Diagnosis: Principles and Recent Advances,” in IEEE Access, vol. 9, pp. 37622-37655, 2021, doi: 10.1109/ACCESS.2021.3062484.
- A. A. Asiri et al., ”Optimized Brain Tumor Detection: A Dual-Module Approach for MRI Image Enhancement and Tumor Classification,” *IEEE Access*, vol. 12, pp. 42868–42887, 2024. doi:10.1109/ACCESS.2024.3379136

References (Part 2)

- O. S. Patil, D. M. Le, E. J. Griffis and W. E. Dixon, ”Deep Residual Neural Network (ResNet)-Based Adaptive Control: A Lyapunov-Based Approach,” 2022 IEEE 61st Conference on Decision and Control (CDC), Cancun, Mexico, 2022, pp. 3487-3492, doi: 10.1109/CDC51059.2022.9992881.

  . Liu, W. Wang and W. He, ”A Study on an Improved ResNet Model for Image Classification,” 2024 17th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (CISP-BMEI), Shanghai, China, 2024, pp. 1-6, doi: 10.1109/CISP-BMEI64163.2024.10906260.
- S. Saifullah, R. Dreżewski, A. Yudhana and A. P. Suryotomo, ”Automatic Brain Tumor Segmentation: Advancing U-Net With ResNet50 Encoder for Precise Medical Image Analysis,” *IEEE Access*, vol. 13, pp. 43473-43489, 2025. doi:10.1109/ACCESS.2025.3547430
- S. Mohsen, A. M. Ali, E.-S. M. El-Rabaie, A. ElKaseer, S. G. Scholz and A. M. A. Hassan, ”Brain Tumor Classification Using Hybrid Single Image Super-Resolution Technique With ResNext101_32×8d and VGG19 Pre-Trained Models,” *IEEE Access*, vol. 11, pp. 55582-55595, 2023. doi:10.1109/ACCESS.2023.3281529
- M. A. Ottom, H. A. Rahman and I. D. Dinov, ”Znet: Deep Learning Approach for 2D MRI Brain Tumor Segmentation,” *IEEE Journal of Translational Engineering in Health and Medicine*, vol. 10, pp. 1-8, 2022, Art no. 1800508. doi:10.1109/JTEHM.2022.3176737

Thank You!